2. Modules & Library

🔹 2. MODULES

📌 What is a Module?

A module is a file containing Python code (functions, variables).

📌 Types

- Built-in (`math`, `random`)
- User-defined

3.1 Importing a Module

python

```
import math
```

```
print(math.sqrt(25))   # 5.0
print(math.pi)         # 3.14159...
```

3.2 from...import

python

```
from math import sqrt, pi
```

```
print(sqrt(16))  # 4.0
print(pi)        # 3.14159...
```

3.3 Aliasing with as

python

```
import math as m
```

```
print(m.ceil(4.2))   # 5
print(m.floor(4.9))  # 4
```

3.4 Creating Your Own Module

python

```
# Save this as myutils.py
def greet(name):
    return f"Hello, {name}!"

def square(n):
    return n * n
```

python

```
# In your main file
import myutils

print(myutils.greet("Alice"))
print(myutils.square(4))
```

3.5 The name Variable

python

```
# myutils.py
def greet(name):
    return f"Hello, {name}!"

if __name__ == "__main__":
    print(greet("Direct run"))
    # This runs only when executed directly, not when imported
```

3.6 Listing Module Contents with dir()

python

```
import math
print(dir(math))
```

3.7 Creating a Package

python

```
# Folder structure:
# mypackage/
#     __init__.py
#     greetings.py
#     calculator.py

# greetings.py
def hello(name):
    return f"Hi {name}!"

# main.py
from mypackage import greetings
print(greetings.hello("Bob"))
```

3.8 init.py

python

```
# mypackage/__init__.py
from .greetings import hello
from .calculator import add

# Now you can do:
# from mypackage import hello, add
```

3.9 Importing with Wildcards (and why to avoid it)

python

```
# Avoid this in production
from math import *
print(sqrt(9))  # works but pollutes namespace

# Prefer explicit imports
from math import sqrt
```

3.10 Checking Module Location

python

```
import math
print(math.__file__)   # shows where the module file lives
print(math.__doc__)    # shows module documentation
```

Program

1. Use math.sqrt

```
importmath
print(math.sqrt(25))
```

2. Random number

```
importrandom
print(random.randint(1,10))
```

3. Import specific function

```
frommathimportfactorial
```

```
print(factorial(5))
```

4. Alias module

```
importmathasm
print(m.pi)
```

5. Random choice

```
importrandom
print(random.choice([1,2,3]))
```

6. Create module (mymodule.py)

```
defgreet():
return"Hello"
```

Main file:

```
importmymodule
print(mymodule.greet())
```

7. Use datetime

```
importdatetime
print(datetime.datetime.now())
```

8. Random float

```
importrandom
print(random.random())
```

9. OS module

```
importos
```

```
print(os.getcwd())
```

10. Sys module

```
importsys
print(sys.version)
```

11. Reload module

```
importimportlib
importmymodule
importlib.reload(mymodule)
```

12. List module functions

```
importmath
print(dir(math))
```

13. Help function

```
help(math.sqrt)
```

14. Generate random list

```
importrandom
print([random.randint(1,100)for_inrange(5)])
```

15. Shuffle list

```
importrandom
lst= [1,2,3]
random.shuffle(lst)
print(lst)
```

16. Time module

```
importtime
print(time.time())
```

17. Sleep function

```
importtime
time.sleep(2)
print("Done")
```

18. Calendar module

```
importcalendar
print(calendar.month(2026,3))
```

19. Platform info

```
importplatform
print(platform.system())
```

20. Custom module with variable

```
# mymodule.py
x=10
```

```
importmymodule
print(mymodule.x)
```

STANDARD LIBRARY

4.1 os Module

python

`import os`

```
print(os.getcwd())           # current directory
os.mkdir("test_folder")      # create folder
print(os.listdir("."))       # list files
os.rename("old.txt", "new.txt")
os.remove("new.txt")
```

4.2 os.path

python

```
import os

path = "/home/user/documents/file.txt"
print(os.path.basename(path))   # file.txt
print(os.path.dirname(path))    # /home/user/documents
print(os.path.exists(path))     # True/False
print(os.path.splitext(path))   # ('/home/.../file', '.txt')
```

4.3 sys Module

python

```
import sys

print(sys.version)       # Python version
print(sys.platform)      # OS platform
sys.exit(0)              # exit program
```

4.4 math Module

python

```
import math

print(math.gcd(12, 18))      # 6
print(math.pow(2, 10))       # 1024.0
print(math.log(100, 10))     # 2.0
print(math.factorial(6))     # 720
```

4.5 random Module

python

```
import random

print(random.randint(1, 10))          # random int 1-10
print(random.choice(["a", "b", "c"])) # random item
random.shuffle([1, 2, 3, 4])
print(random.random())                # float 0.0 to 1.0
```

4.6 datetime Module

python

```
from datetime import datetime, date

now = datetime.now()
print(now.strftime("%d-%m-%Y %H:%M:%S"))

today = date.today()
print(today)
print(today.year, today.month, today.day)
```

4.7 json Module

python

```
import json

data = {"name": "Alice", "age": 25}

# dict to JSON string
json_string = json.dumps(data, indent=2)
print(json_string)

# JSON string to dict
parsed = json.loads(json_string)
print(parsed["name"])
```

4.8 re Module (Regular Expressions)

python

```
import re

text = "My email is alice@example.com"
match = re.search(r'[\w.-]+@[\w.-]+\.\w+', text)
if match:
    print("Email found:", match.group())
```

4.9 collections Module

python

```
from collections import Counter, defaultdict

words = ["apple", "banana", "apple", "cherry", "banana", "apple"]
count = Counter(words)
print(count)                  # Counter({'apple': 3, ...})
print(count.most_common(2))   # top 2
```

4.10 time Module

python

```
import time

start = time.time()
time.sleep(1)   # pause 1 second
end = time.time()
print(f"Elapsed: {end - start:.2f} seconds")
```